

Relationships Between Two Variables: Cross Tables and Scatter Plots

A beginner-friendly guide, rebuilt from the lesson 6 Excel workbook. Everything before this looked at one variable at a time; this is the first time we look at two together.

What you will be able to do after this guide

1. Say which tool to use when you have two categorical variables, and which when you have two numerical ones.
2. Build and read a cross table, including its totals and its percentages.
3. Turn a cross table into a side-by-side bar chart.
4. Draw and read a scatter plot, and tell a positive, negative or no pattern apart.
5. Explain why this leads on to covariance and correlation.

Part 1: The big idea, two variables instead of one

So far we described a single variable: car brands, ice-cream sales per city, test scores. A natural next question is: **do two variables go together?** For example, does the type of investment someone chooses depend on who the investor is? Do students who read well also write well?

Which tool you use depends on what *kind* of variables you have (remember categorical vs numerical from the previous folder):

Your two variables are...	Tool	Example in this workbook
Both categorical (groups)	Cross table , drawn as a side-by-side bar chart	Investor (A, B, C) and type of investment (stocks, bonds, real estate)
Both numerical (numbers)	Scatter plot	Reading score and writing score of each student

Words you will see (keep this table handy)

Word	Plain meaning
Cross table (contingency table)	A table that counts how many cases fall in each combination of two categories
Cell	One box of the table, for example 'Investor A and bonds'
Row total / column total	The sum along a row, or down a column
Grand total	The sum of everything (bottom-right corner)
Relative frequency	A count divided by the grand total, so it becomes a share (0.20 means 20%)
Side-by-side bar chart	A bar chart with several bars in each group, used to draw a cross table
Scatter plot	A graph where each case is one dot, placed using its two numbers
Pattern	The overall shape the dots make, such as rising, falling or just a cloud

Part 2: Cross tables

Step 1: The situation

The workbook records **915 investments**. For each one we know two things: **who** made it (Investor A, B or C) and **what type** it is (stocks, bonds or real estate). Both are categories, so a cross table fits.

Step 2: Build the table

Put one variable down the side (rows) and the other across the top (columns). In each cell, write how many cases have *both* of those categories. Then add the totals. In Excel the totals are just =SUM().

Type of investment	Investor A	Investor B	Investor C	Row total
Stocks	96	185	39	320
Bonds	181	3	29	213
Real Estate	88	152	142	382
Column total	365	340	210	915

Yellow = one cell | Orange = row totals | Green = column totals | Grey = grand total

Figure 1. The cross table. One cell is highlighted in yellow: Investor A made 181 investments in bonds.

Step 3: Read it

What you look at	Example	What it means
A cell	Row 'Bonds', column 'Investor A' = 181	Investor A made 181 bond investments
A row total	Bonds row = 213	All three investors together made 213 bond investments
A column total	Investor A column = 365	Investor A made 365 investments in total
The grand total	915	The number of investments in the whole table

Always do this check

The row totals must add up to the grand total ($320 + 213 + 382 = 915$), and so must the column totals ($365 + 340 + 210 = 915$). If the two do not agree, a number was typed or counted wrongly.

Step 4: Turn counts into shares (relative frequencies)

Counts depend on how big the survey was. Shares are easier to compare. Divide **every number** by the grand total (915). For example, Investor A's bonds: $181 / 915 = 0.198$, which rounds to **0.20**, so 20% of all investments are 'Investor A, bonds'.

Type of investment	Investor A	Investor B	Investor C	Row total
Stocks	0.10	0.20	0.04	0.35
Bonds	0.20	0.00	0.03	0.23
Real estate	0.10	0.17	0.16	0.42
Column total	0.40	0.37	0.23	1.00

The last cell is 1.00 (100%). Real estate is the biggest share of all investments (0.42), and Investor A made the most investments overall (0.40).

Step 5: What does the table tell us? Look within each investor

To see whether the two variables are related, ask: **does each investor split their money in the same way?** Divide each column by *its own* total. This extra step is not in the workbook, but it is the clearest way to read the table:

Share of each investor's investments	Investor A	Investor B	Investor C
Stocks	26.3%	54.4%	18.6%
Bonds	49.6%	0.9%	13.8%
Real estate	24.1%	44.7%	67.6%
Total	100%	100%	100%

The three investors are clearly different. **A** puts half of their money in bonds. **B** almost never uses bonds (just 3 cases) and prefers stocks and real estate. **C** is mostly in real estate. If the type of investment did not depend on the investor, all three columns would look about the same. Because they do not, the two variables are **related**.

Step 6: Draw it as a side-by-side bar chart

A cross table is often drawn as a side-by-side bar chart (called a *clustered column chart* in Excel). Each investor gets a group of bars, one bar per type of investment. Same data, easier to see at a glance:

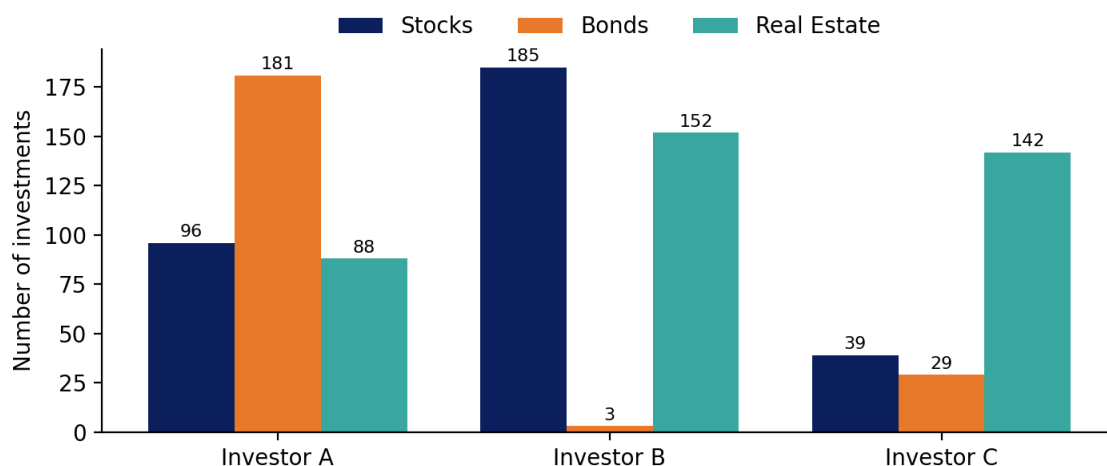


Figure 2. Side-by-side bar chart of the cross table. Compare the heights inside each group.

Making it in Excel

1. Select the table (the counts, with the labels, but not the totals). 2. Insert, then Charts, then Clustered Column. Choosing more than one series makes Excel draw side-by-side bars automatically.

Part 3: Scatter plots

Step 1: The situation

The workbook lists **104 students**. Every student has **two numbers**: a reading score and a writing score. Both are numerical, so we use a scatter plot. Here are the first five students:

Student	Reading	Writing
1	273	216
2	292	282
3	219	250
4	241	217
5	284	266

Step 2: How a scatter plot is drawn

Each student becomes **one dot**. The reading score decides how far right the dot goes (x-axis), and the writing score decides how high it goes (y-axis). Student 1 (reading 273, writing 216) is one dot at that spot. Repeat for all 104 students and you get the picture below.

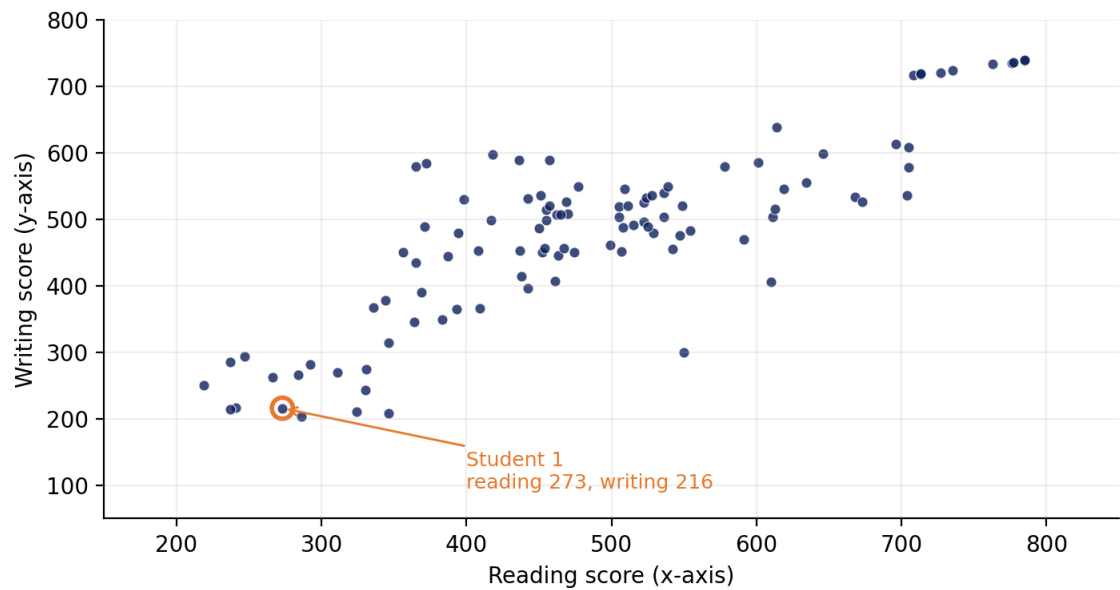


Figure 3. Reading vs writing scores for 104 students. Each dot is one student. The orange circle is Student 1.
(The workbook’s own chart plots only the first 96 students. These notes plot all 104.)

Step 3: How to read a scatter plot

Look at...	In our picture
Direction: do the dots rise, fall, or neither?	They rise from left to right
Strength: are the dots tight around a line, or a loose cloud?	Fairly tight: they follow a clear upward band
Odd ones out: any dots far away from the rest?	A few scattered ones, but most follow the band

What it tells us: students with a higher reading score tend to have a higher writing score too. Notice the word **tend**: it is a pattern across many students, not a rule for every one. Also, it does not say that reading *causes* better writing, only that the two move together.

Step 4: Three patterns to recognise

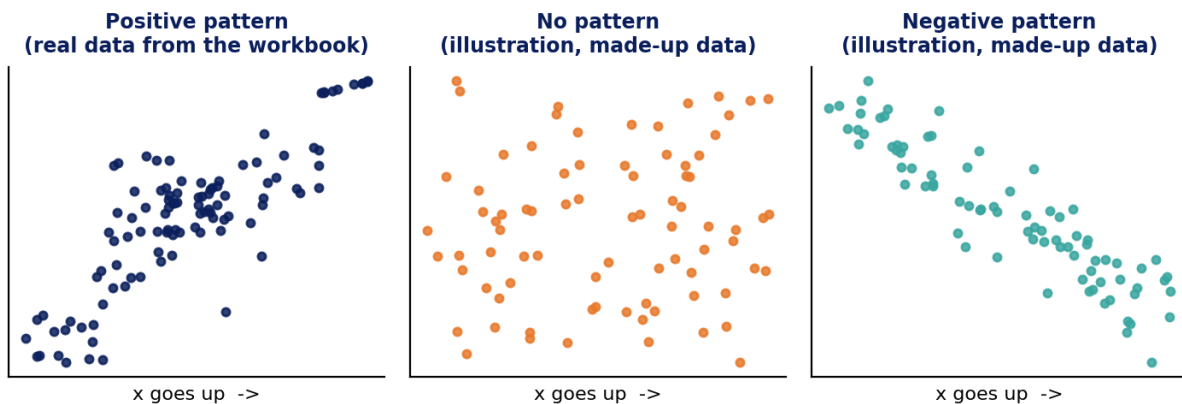


Figure 4. Rising dots = positive. A shapeless cloud = no pattern. Falling dots = negative. The last two panels use made-up data just for illustration.

No pattern means knowing x tells you nothing about y . In the course notes this shows up as dots in a shapeless blob, or a vertical stripe where y changes while x stays the same.

Making it in Excel

1. Select the two columns of numbers (here Reading and Writing). 2. Insert, then Charts, then Scatter. Do not include the Student ID column: it is just a label, not a measurement.

Part 4: Where this leads, putting a number on it

A scatter plot shows a relationship by eye. The next topics in this course measure it with a number: **covariance** and the **correlation coefficient (r)**. Correlation always lies between -1 and +1: close to +1 means a strong rising pattern, close to 0 means no pattern, close to -1 means a strong falling pattern.

For this workbook's 104 students I calculated $r = 0.84$. This number is not in the workbook, and you will learn how to calculate it in the correlation notes. It confirms what the picture shows: a strong positive relationship.

	Cross table	Scatter plot
Variables	Two categorical	Two numerical
What you record	A count in each combination	One dot per case
Question it answers	Do the categories go together?	Do the numbers move together?
Related number later	Shares / percentages	Covariance and correlation

Part 5: Mistakes to avoid

Mistake	Fix
Mixing up row and column totals	Check both add up to the grand total
Comparing raw counts between investors of different size	Use shares within each column, as in Step 5
Using a scatter plot for categories	Scatter plots need numbers on both axes; use a cross table for categories
Including the ID column in the scatter plot	Select only the two measurement columns
Saying 'x causes y' because the dots rise	A pattern shows the variables move together, not why

Part 6: Try it yourself

Practice 1: build a cross table

50 people were asked two things: are you a morning or a night person, and do you drink coffee? Results: morning people: 12 drink coffee, 18 do not. Night people: 15 drink coffee, 5 do not.

Build the cross table with totals, then work out the share of coffee drinkers within each group.

	Drinks coffee	Does not	Row total
Morning person	12	18	30
Night person	15	5	20
Column total	27	23	50

Coffee drinkers: $12 / 30 = 40\%$ of morning people, but $15 / 20 = 75\%$ of night people. The shares differ a lot, so being a night person and drinking coffee seem to go together (related).

Practice 2: which pattern would you expect?

Say whether each pair would probably show a positive pattern, a negative pattern, or no pattern:

(a) hours studied and exam score (b) hours of sleep lost and alertness next day (c) shoe size and exam score

Answers: (a) positive, more study usually means a higher score. (b) negative, less sleep usually means lower alertness. (c) no pattern, the two have nothing to do with each other.

The whole topic in 5 lines

- Two variables together answer: **do they go together?**
- Two categorical variables: **cross table** (counts in each combination, plus totals), drawn as side-by-side bars.
- Divide by the grand total for shares, or by a column total to compare groups fairly.
- Two numerical variables: **scatter plot**, one dot per case. Read direction, strength and odd ones out.
- A pattern means 'move together', not 'one causes the other'. Correlation will put a number on it.